

Artificial Intelligence and Machine Learning in Smart Production: Progress, Trends, and Future Directions

TAMIL R

B.Tech Artificial Intelligence and Data Science
PGP College of Engineering and Technology
Namakkal, Tamil Nadu.
tamil15032007@gamil.com

MANOJKUMAR V

B.Tech Artificial Intelligence and Data Science
PGP College of Engineering and Technology
Namakkal, Tamil Nadu.
manojkumar2604007@gmail.com

PRAVIN G

B.Tech Artificial Intelligence and Data Science
PGP College of Engineering and Technology
Namakkal, Tamil Nadu.
pravin1712007@gmail.com

DINESH R

B.Tech Artificial Intelligence and Data Science
PGP College of Engineering and Technology
Namakkal, Tamil Nadu.
dineshsakthivel2007@gmail.com

Abstract - Adaptation and innovation play a crucial role in advancing the manufacturing industry toward sustainable practices through cutting-edge technologies. To truly foster sustainability, smart production demands a global perspective on applying these technologies. In this context, intensive research in artificial intelligence (AI) has led to established AI-based techniques—like machine learning (ML)—that are already transforming industry for sustainable outcomes.

The aim of this study is to systematically analyse scientific literature on the application of AI and ML in industry. With the advent of Industry 4.0, AI and ML have emerged as the driving forces behind the smart factory revolution. This review classifies the literature by publication year, authors, scientific sector, country, institution, and keywords. We conducted the analysis using the Web of Science and Scopus databases, supported by UCINET and NVivo 12 software.

Additionally, we reviewed empirical studies on ML and AI from the past century to trace the topic's evolution before and after Industry 4.0, spanning 1999 to the present. Out of the literature examined, 82 articles were selected, reviewed, and classified. A key finding is the dominance of publications from the USA, alongside a surge in interest following the introduction of Industry 4.0.

I INTRODUCTION

Modern manufacturing systems require innovative solutions that can improve product quality and make production more sustainable while also lowering operational costs. In recent years, Artificial Intelligence (AI) has played an important role in achieving these goals. AI technologies, supported by Industry 4.0 (I4.0) key technologies such as the Internet of Things (IoT), cloud computing, big data, embedded systems, cognitive computing, and virtual or augmented reality, are transforming traditional industrial systems and creating new models for smart manufacturing.

The concept of artificial intelligence was first clearly defined by John McCarthy, who is widely known as the father of AI. In the 1990s, he described AI as *the science and engineering of creating intelligent machines, particularly intelligent computer programs*. In simple terms, AI refers to machines or computer systems that can perform tasks that normally require human intelligence, such as learning, reasoning, and solving problems.

Artificial intelligence covers a wide range of research areas. Researchers generally classify AI into several major domains, including reasoning, programming, artificial life, belief revision, data mining, distributed AI, expert systems, genetic algorithms, knowledge representation, machine learning, natural language processing, neural networks, theorem proving, constraint satisfaction, and the theory of computation. These fields together

contribute to the development of intelligent systems capable of solving complex problems.

In the 21st century, AI has become a significant research area across many disciplines. It is widely applied in engineering, science, healthcare, education, business, finance, marketing, economics, law, and even stock market analysis. With the rapid growth of machine learning technologies, intelligent machines are increasingly influencing businesses, governments, and society as a whole. Additionally, AI is playing a vital role in promoting global sustainability.

Artificial intelligence can address several critical challenges in sustainable manufacturing. For example, it can optimize energy usage, improve logistics operations, enhance supply chain management, and reduce industrial waste. Because of these advantages, many industries are integrating AI technologies into green manufacturing practices to comply with strict environmental regulations and sustainability goals.

According to sustainability experts, AI has the potential to drive major changes in the way industries operate. In fact, AI is often considered one of the most powerful technologies supporting the Fourth Industrial Revolution, helping organizations achieve more efficient and environmentally friendly production systems.

Several subfields of AI, such as machine learning, natural language processing, image processing, and data mining, have attracted significant attention from both researchers and major technology companies. The rapid development of these technologies has generated strong interest within the scientific community.

Among these subfields, machine learning (ML) has experienced particularly rapid growth. ML techniques allow machines to learn patterns from data and make predictions or decisions without being explicitly programmed. Today, machine learning applications are used in many fields including smart manufacturing, medical diagnosis, pharmaceutical research, agriculture, archaeology, gaming, and business analytics.

Based on these developments, this study conducts a systematic literature review of research related to artificial intelligence and machine learning published between 1999 and 2019. The objective is to create a classification framework that organizes

research articles addressing both AI and ML. This classification helps provide a clearer understanding of how these technologies have evolved and how they are applied across different sectors.

To obtain deeper insights, this study also examines additional factors such as the main research areas and the industrial sectors where AI technologies have the greatest impact. The main contribution of this work is that it provides a comprehensive overview of the existing research on AI and machine learning.

Although many studies have discussed research methods and theoretical approaches in this field, there is still limited integration and comparison between different studies. Therefore, this paper attempts to establish a common perspective on AI and machine learning research and highlight the variations within this domain.

This study does not aim to create a complete framework covering all AI and ML literature. Instead, it serves as a starting point for integrating existing knowledge and identifying potential directions for future research. The paper also explores AI applications in emerging areas such as environmental monitoring, healthcare, maintenance systems, and industrial manufacturing.

Further research is necessary to expand the current understanding of artificial intelligence by combining ideas and theories from traditional disciplines with modern AI frameworks. Such interdisciplinary approaches could lead to more advanced and effective intelligent systems.

The goal of this research is not simply to expand an already well-developed field, but to provide useful insights that may help researchers refine their work and discover new research opportunities. This paper offers valuable perspectives for future studies related to artificial intelligence and machine learning.

Ultimately, the study aims to anticipate how AI research may evolve in the future. As new generations of researchers contribute to this field, the direction of AI development may continue to change and expand. Since this work is a literature review, it establishes a foundation for future investigations and encourages new research questions.

II METHODOLOGY

This study uses a combination of bibliometric analysis, content analysis, and social network analysis to examine research trends. The data for the study was collected from two well-known academic databases: Scopus and Web of Science. Publications between 1999 and 2019 were selected in order to observe how research interest evolved over time, especially before and after the emergence of Industry 4.0.

A systematic literature review approach was adopted to organize and evaluate the selected studies. The overall research process was divided into three major phases.

Phase 1: Research and Classification

This phase involved three main steps. First, relevant publications were identified and bibliometric data was gathered from the selected databases. Next, the collected records were screened to remove unrelated documents and retain only those connected to the research topic.

Phase 2: Data Analysis

After selecting the relevant publications, bibliometric techniques were applied to analyse the dataset. Two indicators were used in the analysis: the total number of papers (TP), which indicates the number of publications, and the total citations (TC), which represents how often these publications have been cited.

To explore the relationships between research topics, Social Network Analysis (SNA) was applied to the keywords extracted from the articles. SNA is commonly used in social science research to examine connections and interactions among different elements within a network. It has been applied in many areas such as trade networks, information flow, institutional studies, and organizational analyse network structure. These tools helped illustrate the relationships between different research themes. In addition, Microsoft Excel was used for data entry and organization.

Furthermore, NVivo 12 software was employed for qualitative data analysis. This tool supported the identification of patterns and relationships among keywords across the selected publications.

Phase 3: Discussion and Interpretation

In the final phase, the results obtained from the analysis were interpreted and discussed. Based on these findings, conclusions were drawn regarding the research trends and key themes related to the study topic. Using this analysis, conceptual connections between different researches themes were identified, which helped in developing interpretative.

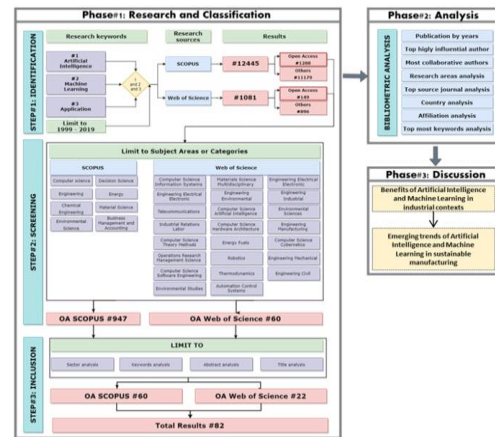


Figure 1. Process flow chart.

III RESULTS OF THE BIBLIOMETRIC ANALYSIS

Phase 1: Data Collection and Classification

The first stage of the study focused on gathering relevant academic publications. This phase involved identifying and collecting research materials from major scientific databases. To organize the process effectively, this stage was divided into three main steps.

1 Identification (Step 1)

To obtain a broad overview of the research topic, a systematic search was conducted using two widely recognized academic databases: Scopus (SCP) and Web of Science (WoS). The search process used Boolean operators to combine key terms related to the research topic. Specifically, the query included the keywords “artificial intelligence” AND “machine learning” AND “application”, as presented in Table 1.

To ensure consistency in the data collection process, the same search keywords were applied across both databases. Additionally, the study considered publications within a 20-year time frame (1999–2019) in order to observe the evolution and growth of research in this area.

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Table 1. Keywords and time period.

Keywords	Time Period
Artificial Intelligence	1999–2019
Machine Learning	
Application	

The initial search process produced a total of 13,512 publications across the selected databases. A comparison of the results indicates that Scopus returned a significantly higher number of documents than the Web of Science (WoS) database. Specifically, the Scopus search identified 12,445 publications, while the Web of Science database yielded 1,081 publications, as summarized in Table

Table 2. Total results of research on Scopus and WoS.

Research Carried out on 2019		
Source of research	Scopus	Web of Science
Results	12,445	1081

The difference in the number of retrieved documents was not surprising. This variation can be explained by the broader coverage of the Scopus database, which is developed by Elsevier and integrates data from several other sources, including platforms such as ScienceDirect and databases indexed through the Scopus search engine research efforts began exploring the possibility of developing machines capable of performing tasks that require human-like intelligence.

In comparison, the Web of Science (WoS) database generally includes a more selective collection of publications, resulting in a smaller number of retrieved records.

In contrast, the results from the Web of Science database indicate a different distribution. In this database, journal articles constitute the largest share (46.12%), while proceedings papers account for 42.86% of the total publications.

A detailed breakdown of all document types identified in the analysis is presented in Table 3.

The difference in the number of documents retrieved from the two databases was expected and can be explained by the different coverage policies adopted by each indexing platform. Scopus, which is developed and maintained by Elsevier, is known for its extensive coverage of scientific literature.

During the following decades, especially from the 1950s to the early 1970s, the field experienced a period of strong optimism, as researchers believed

that rapid progress would soon lead to intelligent machines capable of solving complex problems. funding for AI research declined significantly

Following the growing adoption of these emerging technologies, researchers have increasingly focused on studying and analysing developments associated with Industry 4.0. As the implementation of advanced digital systems continues to expand across industries, and the field entered a period commonly referred to as the “AI winter.” Fig: Research growth on Scopus and Web of Science the academic community has also intensified its research efforts to keep pace with the rapid technological evolution and to explore the potential applications and impacts of Industry 4.0 technologies.

Table 3. Distribution of document types in Scopus and Web of Science.

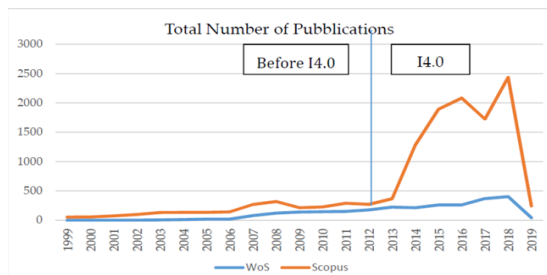
Web of Science			Scopus		
Document Types	Records	Contribute %	Document Types	Records	Contribute %
Article	481	46.12	Conference Paper	7128	57.28
Proceedings paper	447	42.86	Article	4212	33.85
Review	133	12.76	Review	412	3.31
Editorial material	16	1.53	Article in Press	194	1.56
Meeting abstract	2	0.19	Book Chapter	177	1.42
Book chapter	1	0.1	Conference Review	177	1.42
Retracted publication	1	0.1	Book	90	0.72
-	-	-	Editorial	27	0.22
-	-	-	Note	10	0.08
-	-	-	Letter	9	0.07
-	-	-	Short Survey	9	0.07

Screening (Step 2)

The second stage of the research process involved the screening of the collected documents in order to obtain a clearer overview of the relevant research topics and their interdisciplinary connections.

During this phase, specific filtering criteria were applied to refine the dataset including limited computational power, insufficient data availability, and immature algorithmic techniques. As a result, many of the early expectations could not be achieved within the anticipated timeframe. Consequently, and focus only on publications that were most relevant to the objectives of the study.

First, the analysis was limited to open access publications, allowing the researchers to examine and review the full content of the selected documents without access restrictions. Therefore, publications that required subscription or restricted access were excluded from the dataset. This filtering process significantly reduced the number of available documents. As a result, the dataset was narrowed down to 1,288 publications from the Scopus database and 149 publications from the Web of Science (WoS) database.



In the subsequent stage of the research process, an additional level of refinement was introduced through the application of a thematic filtering criterion. This step was specifically designed to ensure that the selected body of literature was closely aligned with the core objectives and scope of the study. Rather than considering all previously identified publications, the researchers focused only on those that fell within clearly defined thematic research areas relevant to the subject under investigation.

Following the implementation of this filtering criterion, as summarized in Table 4, there was a significant reduction in the number of selected documents. The dataset was refined to include 947 publications from Scopus and 60 publications from Web of Science. This reduction indicates the effectiveness of the thematic filtering process in narrowing down a broad initial collection of studies to a more precise and contextually relevant subset.

In essence, this step played a crucial role in ensuring that the final dataset comprised only It should be noted that the number and type of filters applied during the search process vary between the two databases. Although both databases provide similar search functionalities, there are some structural differences in how filtering options are organized. In particular, when considering the those publications that meaningfully contributed to the research objectives, thereby facilitating a more accurate and insightful analysis.

Table 4. Subject area filter on Scopus and WoS.

Subject Area				
Scopus		Web of Science (WoS)		
Computer Science	Chemical Engineering	Computer Science	Artificial Intelligence	Automation Control Systems
Engineering	Energy	Materials Science	Environmental Sciences	Environmental Studies
Materials Science	Decision Science	Engineering Electrical	Computer Science	Operations Research
Environmental Science	Business Management and accounting	Electronic	Hardware Architecture	Management Science
		Telecommunications	Industrial Relations Labor	Robotics
		Engineering Environmental	Engineering Manufacturing	Thermodynamics
		Engineering Industrial	Computer Science Theory Methods	Energy Fuels
		Engineering Civil	Engineering Mechanical	Computer Science Cybernetics
		Computer Science Software Engineering	Multidisciplinary Sciences	

Thematic or subject areas, the Web of Science (WoS) database offers a more detailed and extensive classification system compared to Scopus. As a result, the filtering process related to subject categories is more comprehensive and structured in Web of Science, while Scopus provides a comparatively simpler categorization framework.

Inclusion (Step 3)

After completing the screening stage, the inclusion phase was carried out to determine the final set of documents to be used for the bibliometric analysis. This step involved selecting the most relevant publications from the filtered dataset obtained in the previous stage.

During this phase, the full text of each document was carefully review

This screening process helped to create a more focused and manageable dataset, allowing for a more accurate bibliometric analysis of research trends related to Industry 4.0 and artificial intelligence applications. To evaluate its relevance and eligibility for the study. Each article was independently examined in order to determine whether it contributed meaningful insights to the research topic. In particular, the evaluation focused on identifying studies that demonstrated clear academic relevance and practical significance.

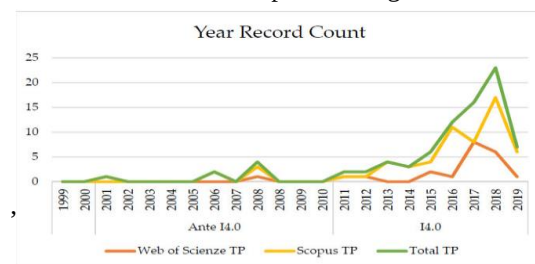


Figure 3. Years of publications.

The selected publications were assessed based on several criteria, such as whether the study presented practical case studies or real-world applications, proposed new Artificial Intelligence (AI) or Machine Learning (ML) algorithms, or discussed potential future developments and research directions in the field. Only those documents that satisfied these criteria were retained for the final analysis.

As a result of this inclusion process, the final dataset used for the bibliometric study consisted of 60 documents obtained from the Scopus database and 22 documents from the Web of Science (WoS) database.

Phase 2: Analysis

This section focuses on the analysis and interpretation of the results obtained from the selected publications.

The purpose of this phase is to examine the collected data and highlight the main patterns, trends, and insights identified through the bibliometric review.

Initially, a general overview of the selected studies is provided in order to summarize the characteristics of the final dataset. This overview includes key information about the publications, such as their distribution, research focus, and overall contribution to the field.

Following this introductory overview, the results of the review are discussed in greater detail. The findings are presented according to the specific research criteria established for the study, with each criterion analysed separately in the corresponding subsections. This structured approach allows for a clearer understanding of the different aspects of the research topic and facilitates a more systematic discussion of the results.

1 Top Highly Influential Analysis

This section identifies the publications that have received the highest number of citations in the Web of Science and Scopus databases.

The list of influential studies is organized according to several attributes, including the research source, publication year, article title, authors, source journal or conference, and the total number of citations recorded in the respective database. The complete list of these highly cited documents is provided in Appendix A.

An examination of Appendix A indicates that the study conducted by José A. Larrañaga and colleagues in 2006 has the highest citation count, reaching 298 citations. This publication focuses on the application of machine learning techniques in bioinformatics and discusses several modelling approaches used for analysing biological data. Since this work was published prior to the emergence of Industry 4.0, it has had a longer period to circulate within the academic community, which has contributed to its high citation count. The article is considered highly influential because it introduced important modelling techniques and contributed significantly to early research developments in the field of computer science.

In general, publications produced before the introduction of Industry 4.0 tend to accumulate higher citation counts, mainly because they have been available for a longer period of time. Nevertheless, it is important to note that several recent publications have also received a considerable number of citations, despite their relatively recent publication dates. This indicates a growing level of interest and engagement from the scientific community in research related to artificial intelligence and machine learning.

The citation analysis also shows that the first highly cited publication within the Industry 4.0 era appeared in 2016. This study, conducted by Bartłomiej Krawczyk, focuses on the development of application models for imbalanced learning problems.

The work emphasizes the need for computationally efficient, adaptive, and real-time learning approaches, while also providing recommendations for future research directions in this area. This article has received 119 citations, reflecting its significant impact on the research community.

Another notable contribution is the work published by Thorsten Wuest and colleagues, which has also attracted considerable attention from researchers.

This publication offers comprehensive overview of various machine learning techniques and discusses their potential applications across different domains.

Overall, the citation analysis reveals that the average number of citations per document in the dataset is approximately 16.58. Considering the rapid growth of research in Artificial Intelligence (AI) and Machine Learning (ML), it is expected that this average citation value will continue to increase in the coming years as more studies build upon the existing body of knowledge.

2 Publications by Year

In accordance with the observations presented in Section 3.1.1, the analysis indicates that the number of publications included in the dataset remained relatively low during the years preceding the emergence of Industry 4.0. However, a noticeable increase in the number of publications can be observed starting from 2012, reflecting the growing academic and industrial interest in advanced digital technologies.

The trend illustrated in Figure 3 highlights

3 Most Collaborative Authors

The analysis shows that the majority of the selected publications were produced through collaborative research efforts, involving more than one author. This indicates that research in fields such as Artificial Intelligence (AI) and Machine Learning (ML) often requires interdisciplinary collaboration among researchers with different areas of expertise.

From this perspective, it was possible to determine the number of contributing authors for each publication included in the dataset. The results, illustrated in Figure 4, reveal that most of the studies

were written by research teams consisting of two to five authors.

To evaluate the level of collaboration among researchers, specific bibliometric indicators were used that research output related to this topic was limited during the earlier years of the study period. In particular, there are two intervals—2001 to 2008 and 2008 to 2011—where the number of publications appears to be minimal. These gaps suggest that, during those periods, the technological applications associated with artificial intelligence and machine learning were still in relatively early stages of development and had not yet become widely adopted in industrial contexts.

One of the primary indicators applied in this analysis is Total Papers (TPs), which represents the overall number of publications produced by each author or group of authors within the dataset

This metric helps identify the most active contributors and provides insight into the collaborative structure of the research community.

From this perspective, it was possible to determine the number of contributing authors for each publication included in the dataset. The results, illustrated in Figure 4, reveal that most of the studies were written by research teams consisting of two to five authors.

4 Research Areas Analysis

The analysis of research areas was conducted using the 82 selected publications included in the final dataset. During this process, a total of 164 research area classifications were identified. This number is higher than the total number of papers because a single publication can be associated with multiple research disciplines depending on its scope and methodology.

Due to the limited number of publications identified in the years preceding the emergence of Industry 4.0, the distribution of research areas mainly reflects developments that occurred during the period associated with the current industrial revolution. The findings therefore highlight how the introduction of the industry 4.0 paradigm has significantly accelerated both technological adoption and scientific research activities

Among the various disciplines represented in the dataset, computer science emerges as the most prominent research area, accounting for approximately 29% of the total publications. This is followed by engineering, which contributes around 23%, and biochemistry, genetics, and molecular

biology, which together represent about 6% of the publications. These results indicate that the application of artificial intelligence and machine learning technologies is strongly

In addition to these primary areas, several other disciplines were identified. Many of these fields are considered interdisciplinary or complementary to the leading research areas. This interdisciplinary nature can largely be attributed to the influence of Industry 4.0, which encourages the integration of advanced technologies across multiple sectors and research domains connected with computational and engineering-related fields, while also extending into life sciences and other scientific domains.

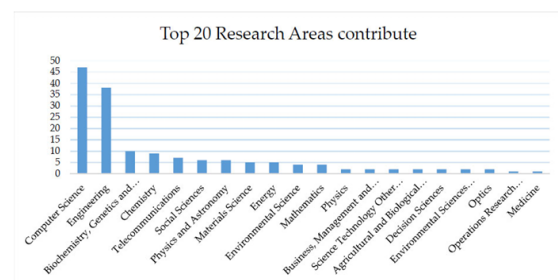


Figure 5. Top 20 research areas contributions.

When considering the percentage contribution, the top three research areas collectively account for nearly 60% of the total publications included in the analysis. Furthermore, when examining the top 20 research areas, the distribution presented in Figure 5 shows a strong concentration within the major disciplines previously mentioned. In fact, the first five research areas alone represent approximately 70% of the analysed publications, demonstrating a clear dominance of certain fields in the development of AI and machine learning research.

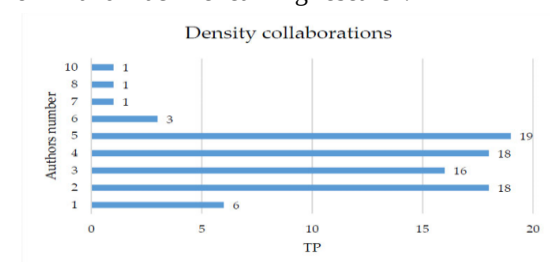


Figure 4. Collaborative groups.

Overall, these findings confirm that AI and machine learning research attracts significant interest across a wide variety of scientific and technological fields, demonstrating the widespread impact and relevance of these technologies in the modern research landscape.

5 Top Source Journals Analysis

This section examines the most frequently used publication sources, identifying the top journals and

conference proceedings where the selected studies were published. Academic journals play an important role in the research community, as they provide a structured platform for disseminating scientific knowledge and tracking developments within specific disciplines.

In the dataset analysed for this study, a total of 74 different publication sources were identified from the selected documents. However, when focusing on the top 20 most frequent sources, the distribution shows that only 13 sources have published more than one article related to the research topic. Together, these 13 sources represent approximately 43% of the total publications, indicating that the remaining papers are distributed across a wide range of other journals and conference proceedings.

When the sources from the two databases were analysed independently, differences in the rankings were observed. Within the Web of Science dataset, the journal IEEE Access appears as the most frequent source, with two publications. In contrast, the results obtained from the Scopus dataset indicate that the leading sources are Procedia Computer Science, MATEC Web of Conferences, and Machine Learning, each containing four publications, representing approximately 5% of the total dataset.

When the results from both databases are combined, the overall ranking becomes more aligned with the results observed in the Scopus dataset. In the aggregated analysis, the journals

Procedia Computer Science, MATEC Web of Conferences, and Machine Learning appear as the most frequent publication sources. Consequently, IEEE Access, which ranked first in the Web of Science dataset, moves down to approximately the eighth position in the combined ranking.

Beyond these leading sources, around ten additional journals contribute approximately 3% of the total publications each, while the remaining sources contribute only one publication each, corresponding to about 1% per source.

The relatively low concentration of publications within a small number of journals suggests that research related to artificial intelligence and machine learning applications is widely distributed across various scientific outlets. This indicates a broad and interdisciplinary interest in the topic across multiple academic communities.

As expected, some specialized journals focusing on artificial intelligence, such as AI-related publications and machine learning journals, are among the most frequent sources. However, it is also noteworthy that several journals from other scientific disciplines contribute to the literature. Examples include Sustainability, BMC Bioinformatics, and Nuclear Engineering and Design, which demonstrates the multidisciplinary nature of AI and ML applications.

The relatively low concentration of publications within a limited number of sources suggests that research in the field of artificial intelligence (AI) and machine learning (ML) is highly diverse and interdisciplinary. Instead of being restricted to specialized AI journals alone, studies are published across a wide variety of scientific outlets. This reflects the broad applicability of AI and ML techniques across multiple domains.

In conclusion, the analysis of publication sources reveals that although a small number of journals and conferences play a key role in disseminating research, the overall landscape is highly distributed and multidisciplinary.

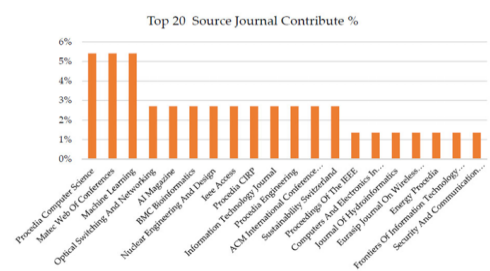


Figure 6. Top 20 source journals contributions.

6 Country Analysis

The country-level analysis conducted using data from both Scopus and Web of Science reveals consistent results regarding the geographical distribution of research contributions. In both databases, China and the United States emerge as the leading contributors to scientific publications related to artificial intelligence and machine learning applications. The distribution of these contributions is illustrated in Figure 7.

This outcome is not surprising, considering that these two countries have very large populations and extensive research infrastructures. With populations exceeding one billion in China and over three hundred million in the United States, the number of researchers and academic institutions involved in technological research is considerably higher than in many other regions. Consequently, these countries

are able to produce a significant portion of global scientific output in advanced technological fields.

When focusing specifically on Europe, Germany appears as the most productive



Finally, Figure 6 illustrates the contribution of the top 20 publication sources, highlighting the distribution of research output among growing importance and versatility of AI and ML technologies across different scientific and academic communities. Country in terms of publications within the analysed dataset. This result is consistent with the origins of the industry 4.0 concept, which was initially developed in Germany as part of a national strategy to promote digital transformation in manufacturing and industrial systems.

However, the overall distribution of research contributions suggests that China and the United States dominate the global landscape, while European countries collectively contribute fewer publications compared to these leading nations. This observation highlights an ongoing debate regarding Europe's position in the global competition for technological leadership, particularly in the field of artificial intelligence.

In discussions about the future of AI research and innovation, Cédric Villani emphasized the importance of strengthening Europe's capabilities in artificial intelligence. According to his perspective, Europe must enhance its competitiveness with global leaders such as China and the United States while also ensuring that technological development aligns with ethical principles and societal values. AI development is increasingly viewed as an area of strategic and geopolitical competition, with the potential to influence global technological leadership in the coming decades despite the current gap in research output, European countries have recently taken steps to strengthen cooperation in digital technologies. Since 2017,

7 Affiliation Analysis

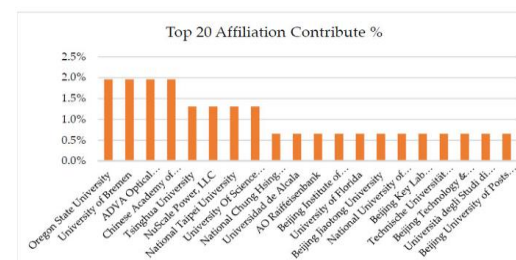
The affiliation analysis was conducted to identify the universities, research institutes, and organizations that contributed to the publications included in the dataset. From the 82 selected papers, a total of 153 different institutional affiliations were identified. This number is higher than the total number of papers because many publications involve collaboration between multiple institutions.

When focusing on the top 20 affiliations, the distribution indicates that most institutions are associated with only one publication, showing a one-to-one relationship between the paper and the corresponding affiliation. Among the most active institutions, only four affiliations contributed three publications

As these collaborative initiatives continue to expand, it is expected that Europe will production systems and advanced industrial technologies countries such as France, Germany, and Italy have intensified their collaboration efforts to accelerate the digital transformation of the manufacturing sector. These initiatives aim to promote the development of smart experience significant progress in smart manufacturing and Industry 4.0 technologies, which may also lead to an increase in scientific research and publications in the near future. The distribution of research contributions by country is illustrated in Figure 7, which highlights the relative participation of different nations in this research area. Each, representing approximately 2% of the total contribution, while another four institutions contributed two publications each, accounting for about 1.3% of the dataset.

Events of this type bring together researchers, policymakers, and industry leaders to discuss technological advancements and future strategies in the field of AI.

Overall, the affiliation analysis demonstrates that research on artificial intelligence and machine learning applications involves a broad network of institutions across different countries and continents. The findings also confirm that collaboration among universities.



8 Top Keywords Analysis

The analysis of keywords was performed to identify the most frequently used terms within the selected publications and to better understand the main research themes emerging from the dataset. For this purpose, the software NVivo 12 was used to extract and analyse the keywords associated with each document. Through this process, the top 20 most frequently occurring keywords were identified. These keywords represent the terms that appear most consistently across the evolving technological field selected publications and therefore reflect the main research focus areas of the analysed literature sizes, while less common terms are shown in smaller text.

From this representation, it can be clearly observed that the most prominent terms are “machine,” “learning,” and “intelligence,” which appear significantly larger than the other words in the visualization. This result is expected, as these terms are central to the research domain analysed in this study.

In addition to the word cloud representation, another visualization technique known as a tree map (or tree words diagram) was used to illustrate the relative importance of the extracted keywords. This representation, shown in Figure 10, organizes keywords into rectangular blocks whose size reflects their frequency of occurrence within the dataset.

As expected, the keyword analysis reveals that the most frequently occurring terms within the dataset are “learning,” “machine,” and “intelligence.” These keywords appear with the highest frequency because they directly represent the core technologies examined in this study, namely artificial intelligence and machine learning. The strong presence of these terms confirms that the selected publications are primarily focused on the development, application, and analysis of intelligent computational systems.

Although it is predictable that technology-related terms dominate the keyword rankings, the analysis also Similar to the word cloud visualization, the largest boxes correspond to the most frequently indexed keywords, highlighting the dominant research themes across the selected publications.

Overall, the keyword analysis provides valuable insights into the primary research topics and thematic trends present in the literature. The prominence of terms related to machine learning and artificial intelligence confirms the central role of these technologies in the analysed studies, while the presence of additional keywords reflects the diverse

This section emphasizes the presence of several additional keywords that are closely associated with a variety of application domains. The identification of these keywords provides important insights into the scope and reach of the research field. Specifically, it suggests that artificial intelligence (AI) and machine learning (ML) are not limited to a single specialized domain; rather, they are widely applicable across numerous disciplines. Their integration within the broader framework of Industry 4.0 has significantly expanded their influence, enabling these technologies to be utilized in diverse sectors such as industrial automation, advanced data analysis, healthcare systems, organizational management, and decision-support mechanisms.

A closer examination of the dataset reveals that certain keywords—such as “data,” “neural,” “decision,” and “management”—appear with moderate to high frequency. The repeated occurrence of these terms indicates that AI and ML techniques are frequently applied in key functional areas. For instance, the term “data” highlights the importance of data processing and analytics, application areas and interdisciplinary nature of research in this field which form the foundation of most AI systems.

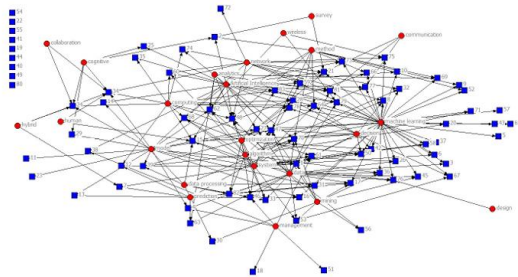
Similarly, the keyword “neural” points to the widespread use of neural networks and deep learning models in solving complex computational problems. The presence of “decision” reflects the role of AI in supporting intelligent decision-making systems, while “management” indicates its application in optimizing organizational processes and resource allocation.

Together, these keywords confirm the interdisciplinary nature of AI and ML research. They demonstrate how these technologies are increasingly being used to address complex analytical challenges and operational tasks across a wide range of domains. This growing integration underscores the transformative impact of AI and ML in both academic research and real-world applications.

Overall, the combination of keyword analysis and social network analysis provides a comprehensive understanding of the thematic structure and interconnectedness within the research domain. These methods reveal how AI and ML research is deeply integrated with multiple application areas and highlight the complexity of technological development highlights the presence of several additional keywords connected to other application

domains. This observation suggests that artificial intelligence and machine learning are not confined to a single research field but instead extend across a wide range of disciplines. Their integration within the framework of Industry 4.0 has enabled these technologies to influence multiple sectors, including industrial automation, data analysis, healthcare, management, and decision-support systems.

More specifically, keywords such as “data,” “neural,” “decision,” and “management” appear with moderate to high frequency in the analysed publications. The presence of these terms indicates that AI and machine learning technologies are frequently applied to areas such as data processing, neural network modelling, decision-making systems, and organizational management processes. This confirms the interdisciplinary nature of AI research and its growing role in supporting complex analytical and operational tasks across different domains within the industry 4.0 paradigm.



In addition to keyword frequency analysis, another analytical method used in this study involves social network analysis (SNA), which was conducted using the software UCINET. Social network analysis is a methodological approach used to study the relationships and interactions between different entities within a network. This technique provides a graphical and mathematical representation of complex systems where multiple elements interact with one another.

Using the UCINET software, a graphical representation of the network relationships among keywords was generated

The resulting visualization, illustrated in Figure 11, represents a sociometric network in which nodes correspond to keywords and connections represent relationships or associations between them. In this network structure, each relationship is typically illustrated by an oriented arrow, indicating the direction of interaction or association within the analysed dataset.

Overall, the keyword and network analyses provide valuable insights into the structure and thematic connections within the research field. They highlight

how artificial intelligence and machine learning research is interconnected with multiple application domains and demonstrate the complexity and interdisciplinary nature of technological development within the industry 4.0 paradigm

SNA has been widely applied in several fields within the social sciences and has more recently been adopted in research areas such as information diffusion, international trade networks, institutional analysis, and organizational structures.

Phase 3: Discussion

Artificial Intelligence (AI) and Machine Learning (ML) play a vital role in modern industrial systems by supporting digital transformation and innovation. These technologies enable organizations to analyse large datasets, improve decision-making, and optimize industrial processes. Their implementation leads to increased efficiency, better resource utilization, reduced production time, and improved product quality.

AI and ML research is globally distributed, with major contributions from countries like the United States and China, which are heavily investing in technological advancement. Furthermore, in the context of sustainable manufacturing, AI and ML help reduce energy consumption, minimize environmental impact, and promote efficient use of resources. Overall, these technologies are key drivers of innovation, industrial growth, and sustainability in the modern era.

IV CONCLUSION

This study examined the current state of research on Artificial Intelligence (AI) and Machine Learning (ML) applications, focusing on their increasing importance in modern scientific and technological developments. Because the amount of literature available on these topics has grown rapidly in recent years, conducting a comprehensive review of all published studies is extremely difficult. For this reason, a systematic and structured approach was adopted to identify and analyse the most relevant publications related to AI and ML.

The study presented a systematic review of AI and ML applications across several scientific domains, highlighting the rapid expansion of research in these fields. To ensure transparency and reliability in the selection process, the documents were chosen using clear and objective research criteria, minimizing the influence of subjective judgments by the researchers. Through this approach, the study aimed

to provide a clear overview of how AI and ML technologies are currently being applied in different research areas.

Another objective of this work was to establish a conceptual framework of the existing literature, which can serve as a reference point for future studies. By analysing the available publications, this research helps researchers better understand the current development trends in AI and ML while also identifying potential directions for future investigation and innovation.

It is important to note that this review was conducted using only two major academic databases, namely Scopus and Web of Science, and it included only open-access publications. As a result, a significant number of relevant studies that are available through restricted access sources or other indexing platforms were not included in the analysis. Future research could therefore expand the scope of the review by incorporating additional databases such as Google Scholar and by considering subscription-based publications to provide a more comprehensive overview of the research landscape.

Overall, the findings of this study confirm the growing importance of AI and ML technologies in both academic research and industrial applications, particularly in areas related to smart manufacturing, sustainability, and digital transformation. As these technologies continue to evolve, they are expected to play an even more significant role in shaping future innovations and scientific developments

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